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2 August 1982
SUPERSEDING
MIL-C-51012A
12 February 1970

MILITARY SPECIFICATION

o-CHLOROBENZALDEHYDE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of o-chlorobenzaldehyde.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
D1193 - Reagent Water
D1493 - Solidification Point of Industrial Organic Chemicals

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. o-Chlorobenzaldehyde shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Assay as o-chlorobenzaldehyde, percent by weight	97.0	---	4.2.4.1
Ash, percent by weight	---	0.02	4.2.4.2
Ethanol insolubles, percent by weight	---	0.1	4.2.4.3
Acidity (as o-chlorobenzoic acid), percent by weight	---	1.0	4.2.4.4
Specific gravity, 25°/25°C	1.245	1.250	4.2.4.5
Freezing point, °C	7.0	---	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the o-chlorobenzaldehyde produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test (see 6.5). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1 liter shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Drums incapable of being stacked
(f) Drums incapable of supporting superimposed loads
(g) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests should be conducted as follows:

4.2.4.1 Assay as o-chlorobenzaldehyde.

(a) Apparatus Use a gas-liquid chromatograph with isothermal or temperature programming. Recommended chromatographic conditions for an F & M Model 720 chromatograph are shown in table III. These conditions may be modified, as necessary, when other chromatographs are used.

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TABLE III. Chromatographic conditions

Characteristic	Condition
Detector	Thermal conductivity
Bridge current, milliamps	135
Attenuation	4X (as needed)
Carrier gas	Helium
Carrier gas flow rate, ml per minute	90
Detector temperature, °C	200
Injection port temperature, °C	220
Programmed column temperature, °C	5° C per minute from 65° to 190°
Column material	pyrex glass
Column dimensions	6 feet by 1/4 inch OD
Column support	60/80 mesh "Gas Chrom Q"
Column coating	10 percent "QF-1"

(b) Calibration. Prepare a 50 milligram (mg) per milliliter (ml) solution of o-chlorobenzaldehyde of known purity in benzene. Inject 5-, 10-, 15-, 20-, 25-, 30-, 35-, 40-, 45-, and 50-microliter quantities separately into the chromatograph and measure the integrated area produced by each quantity. Plot the weights in micrograms versus the respective areas produced. Repeat the procedure using a 50 mg per milliliter solution of an internal standard of known purity in benzene. (1,10-Dibromodecane has been found satisfactory for use as an internal standard; other purified chemicals may be used.) Chromatographic conditions must be identical in both the internal standard and the o-chlorobenzaldehyde of known purity determinations. Plot the weights in micrograms of the internal standard versus the respective areas produced, on the graph containing the plotted o-chlorobenzaldehyde results. Calculate a response factor from the ratio of the slopes of the two plotted lines, corrected for purity, as follows:

$$R = \frac{AB}{CD}$$

where: A = Area per unit weight of the o-chlorobenzaldehyde of known purity,
B = Percent purity of the internal standard,
C = Area per unit weight of the internal standard, and
D = Percent purity of the o-chlorobenzaldehyde.

(c) Procedure. Weight to the nearest milligram approximately 0.5 gram (g) each of the specimen and the internal standard into a 10-ml volumetric flask. Dissolve and dilute to the mark with benzene. Saturate the chromatograph column three times with 20-microliter portions of the solution. Inject a fourth 20-microliter portion of the solution and measure the resulting areas. (Using the conditions in table III, the o-chlorobenzaldehyde will elute in 7.5 minutes at 104°C and the 1,10-dibromodecane will elute in 20 minutes at approximately 165°C.) Calculate the percent o-chlorobenzaldehyde as follows:

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$$\text{Percent o-chlorobenzaldehyde} = \frac{ABC}{DER}$$

where: A = Area produced by the specimen solution,
B = Weight in grams of the internal standard,
C = Percent purity of the internal standard,
D = Area produced by the internal standard solution,
E = Weight in grams of the specimen, and
R = Response factor calculated in (b).

4.2.4.2 Ash. Weigh to the nearest milligram approximately 10 g of the specimen and evaporate it slowly so as not to splatter the specimen (the boiling point is approxmately 208°C). Heat the residue to constant weight in a furnace at a temperature of no less than 650°C. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100\ A}{W}$$

where: A = Weight in grams of residue, and
W = Weight in grams of specimen.

4.2.4.3 Ethanol insolubles. Weight to the nearest 0.1 milligram approximately 1 g of the specimen and dissolve in 20 ml of ethanol. Filter through a tared filter crucible, dry, and weigh to the nearest 0.1 mg. Calculate the percent ethanol insolubles as follows:

$$\text{Percent ethanol insolubles} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of filter crucible and residue,
B = Weight in grams of filter crucible, and
W = Weight in grams of specimen.

4.2.4.4 Acidity. Mix approximately 10 ml of the specimen, measured to the nearest 0.1 ml, with 10 ml of ethanol. Titrate to a violet end point with 0.1N sodium hydroxide solution using phenol red indicator. Calculate the percent acidity as o-chlorobenzoic acid as follows:

$$\text{Percent acidity} = \frac{15.7\ AB}{CD}$$

where: A = Milliliters of sodium hydroxide solution used,
B = Normality of sodium hyroxide solution,
C = Specific gravity determined in 4.2.4.5, and
D = Milliliters of specimen used.

4.2.4.5 Specific gravity. Determine specific gravity at 25°/25°C in accordance with the pycnometer method of ASTM D891.

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4.2.4.6 Freezing point. Determine the freezing point in accordance with ASTM D1493.

5. PACKAGING

5.1 Unit packing, level C. A quantity of 525 (+5 or -0) pounds (lb) of o-chlorobenzaldehyde shall be unit packed level C in accordance with Department of Transportation (DOT) regulations in a nominal 55-gallon steel drum conforming to DOT specification 5, 5A, 5B, 5C, 6A, 6B, or 6C. The drum shall have an inner protective coating which shall neither affect nor be affected by the contents during shipment from the supply source to the first receiving activity and for a period of one year. The drum closures shall be torqued closed in accordance with the drum manufacturer's instructions. The unit pack shall comply with Uniform Freight Classification Rules, National Motor Freight Classification Rules or other carrier rules applicable to the mode of transportation. Drums shall be capable of being stacked and of supporting superimposed loads during shipment and storage, and shall assure safe delivery to destination without leakage or adversely affecting the purity of contents.

5.2 Packing, level C. The o-chlorobenzaldehyde, unit packed as specified in 5.1, shall require no further protection for shipment.

5.3 Marking. Drums of o-chlorobenzaldehyde shall be marked in accordance with DOT regulations and MIL-STD-129. In addition, each drum shall be marked to show the lot or batch number, the date of manufacture of the o-chlorobenzaldehyde, and the following precautionary marking:

CAUTION!

Avoid prolonged or repeated contact with skin.
Do not swallow.

6. NOTES

6.1 Intended use. o-Chlorobenzaldehyde is intended for use in the manufacture of Riot Control Agent, CS.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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6.5 Sampling and testing precautions. o-Chlorobenzaldehyde is an irritating material and a combustible liquid. Avoid inhalation and contact with skin, eyes and clothing. Keep away from heat and open flames.

Custodian:

Army - EA

Review activities:

Army - MD
DLA - GS

User activity:

Army - SM

Preparing activity:

Army - EA

Project No. 6810-B325

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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